Taller 4.2: Transformers y el Mecanismo de Atención

En la primera parte de este taller vimos como puede entrenarse y usarse recurrentes poderosos implementar modelos de deep learning con la capacidad de entender texto escrito y usar esa comprensión para clasificar mensajes en diversas tareas. A continuación profundizaremos una versión alternativa, basada en el mecanismo de atención y en una arquitectura de tipo Sample Transformer, que es actualmente la que ha revolucionado ampliamente el procesamiento y entendimiento del lenguaje natural y a permitido la aparición de los grandes modelos de lenguaje (Large Language Models, LLM) y servicios como ChatGPT. GPT del hecho es un acrónimo de Generative Pre-Trained Transformer.

Comenzaremos con un ejemplo de entrenar un transformer desde cero para ver como funciona internamente y luego usaremos un modelo ya preentrenado disponible para nuestro beneficio.

In [1]:
```
import tensorflow as tf
from keras.models import Sequential, Model
from keras.layers import *
from keras.utils.data_utils import get_file
from keras.preprocessing.text import Tokenizer, text_split
from keras.layers import Dense, Embedding, Input, Attention, PlotNet
import numpy as np
import datetime
import os
from nltk.tokenize import word_tokenize,sent_tokenize,wordpunct_tokenize
```

Out[1]:
```
datetime.I' (mod 'nltk.tokf')
```

In [2]:
```
MAX=
MAXLEN=20
text = open(get_file('cervantes.txt', 'https://raw.githubusercontent.com/fsancho/PLN/master/cervantes.txt')).read()
for line in text.split('\n'):
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    for t in tokens:
        if t not in vocabulario:
            if len(vocabulario)==MAX:
                continue
            else:
                vocabulario[t]=len(vocabulario)
    if len(tokens)>MAXLEN:
        continue
    if len(tokens)>maxlen:
        maxlen=len(tokens)
print("Longitud del vocabulario",len(vocabulario))
print("maxlen",maxlen)
```

Out[2]:
```
MAX
MAXLEN
Longitud del vocabulario 1000
maxlen 20
```

In [3]:
```
from tensorflow.keras.preprocessing.text import Tokenizer

tokenizer = Tokenizer()
tokenizer.fit_on_texts(text)
sequences = tokenizer.texts_to_sequences(t)

vocabulario=tokenizer.word_index
print("Longitud del vocabulario",len(vocabulario))
vocabulario['PAD']=0
vocabulario['<UNK>']=len(vocabulario)
```

Out[3]:
```
Longitud del vocabulario 2693
```

In [4]:
```
X=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp)
X=np.array(X)
print(X[0])
```

Out[4]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [5]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[5]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [6]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[6]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [7]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[7]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [8]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[8]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [9]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[9]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [10]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[10]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [11]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[11]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [12]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[12]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [13]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[13]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [14]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[14]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [15]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[15]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [16]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[16]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [17]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[17]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [18]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[18]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [19]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[19]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [20]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[20]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [21]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[21]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [22]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[22]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [23]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[23]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [24]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[24]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [25]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[25]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [26]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[26]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [27]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[27]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [28]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[28]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [29]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[29]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [30]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[30]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [31]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[31]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [32]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[32]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [33]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[33]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [34]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[34]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [35]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[35]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [36]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[36]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [37]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[37]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [38]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[38]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [39]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[39]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [40]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[40]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [41]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[41]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [42]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[42]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [43]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[43]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [44]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[44]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [45]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[45]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [46]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[46]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [47]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[47]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [48]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[48]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [49]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[49]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [50]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[50]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [51]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[51]:
```
[ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1] [ 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```

In [52]:
```
X=[]
Y=[]
for line in t:
    if len(line)>MAXLEN:
        continue
    tokens=tokenizer.tokenize(line)
    if len(tokens)>MAXLEN:
        continue
    tmp=[]
    for tokens in tokens:
        tmp.append(vocabulario[tokens])
    while len(tmp)==MAXLEN:
        tmp.append(0)
    X.append(tmp[:-1])
    Y.append(tmp[1:])
X=np.array(X)
Y=np.array(Y)
print(X[0],Y[0])
```

Out[52]:
```
[ 1 1 1 1 1 1 1 1 1 
```